



Alternative Synthetic Route to Chiral N-Substituted Camphor-Derived β -Amino Alcohols

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ABSTRACT An alternative route from (1R)-(+)-camphor to chiral N-substituted camphor-derived β -amino alcohol (**4b–e**) consists of four steps with a total yield of 28%. N-Alkylation of camphor-derived β -amino alcohol (**4a**) involves condensation and hydride reduction in one pot without isolation of intermediates. Condensation of **4a** with aldehydes or ketones generates a mixture of 1,3-oxazolidines (**6**) and imino-alcohols (**7**), which are reduced to **4b–e** by NaBH₄. *Chirality* 19:235–238, 2007. © 2007 Wiley-Liss, Inc.

KEY WORDS: β -amino alcohol; camphor; asymmetric synthesis

INTRODUCTION

Most catalytic asymmetric syntheses^{1–4} are catalyzed by transition metals in combination with chiral ligands.^{5–8} Some of the popular chiral ligands, which have been extensively used for this purpose, are β -amino alcohols.^{9–14} Some of the chiral β -amino alcohols are natural products, such as cinchonine, cinchonidine, quinine, quinidine, ephedrine, and norephedrine.^{15–22} Most of the chiral β -amino alcohols were prepared from camphor derivatives,^{23–27} reduction of enantiopure amino acids,^{9,28–30} Sharpless' aminohydroxylation of olefins,^{31,32} Mannich-type reactions for the direct addition of α -hydroxy ketones to imines,^{33–35} regioselective ring opening of chiral epoxides and aziridines,^{36,37} and nucleophilic addition to α -amino carbonyl compounds and α -hydroxy imines.³⁸

Preparation of chiral camphor-derived β -amino alcohols has been of interest because they have been successfully used as chiral auxiliaries and ligands in asymmetric syntheses recently.^{23–27} Thornton²³ prepared chiral camphor-derived β -amino alcohol (**4a**) from condensation of (1R)-(-)-camphorquinone (**1**) with hydroxylamine, followed by hydride reduction with NaBH₄, and hydrogenation with 50 psi of H₂ and PtO₂ catalyst (Scheme 1). Denmark²⁴ made chiral N-substituted camphor-derived β -amino alcohol (**4b–d**) from condensation of (1R)-(-)-camphorquinone (**1**) with primary amines in the presence of TiCl₄, followed by hydride reduction with NaBH₄ (Scheme 2). Tanaka²⁵ synthesized chiral N-alkyl camphor-derived β -amino alcohol (**4e**) from hydride reduction of (1R)-(+)-3-(hydroxyimino)camphor (**2**) with LiAlH₄, followed by condensation with diethyl carbonate, separation of the diastereomers, base-catalyzed hydrolysis, acylation with acid chlorides, and hydride reduction with LiAlH₄ (Scheme 3).

Alkylation of amines with alkyl halides usually suffers uncontrollably alkylation, so at least three strategies were proposed to cope with this problem: (1) Gabriel amine synthesis, (2) reductive amination, (3) conversion

of amines to amides followed by reduction of the amides.³⁹ Literatures in the preparation of N-alkyl-substituted camphor-derived β -amino alcohols usually adopted the method (3).^{25,40,41}

In this article, we followed an alternative route from (1R)-(+)-camphor to N-alkyl-substituted camphor-derived β -amino alcohols (**4b–e**). In this route, we modify hydrogenation of *exo*-hydroxyoxime (**3**) with an easier process and propose a general way for N-alkylation of chiral camphor-derived β -amino alcohol (**4a**). In the N-alkylation reactions, we can do the two processes of reactions in one pot without isolation of intermediates.

MATERIALS AND METHODS

General

All reagents and starting materials were obtained from commercial sources and were used as received.

Synthesis

(1R,2S,3R,4S)-3-*exo*-amino-2-*exo*-hydroxy-1,7,7-trimethylbicyclo[2.2.1]heptane (4a). To a solution of **3** (0.34 g, 1.9 mmol) in 10 mL of dry ethyl ether was added lithium aluminum hydride (0.42 g, 11 mmol). The mixture was stirred in a nitrogen atmosphere at reflux condition for 48 h. After the reaction was complete, an

Contract grant sponsor: NSC, Taiwan; Contract grant number: NSC 94-2113-M-006-009.

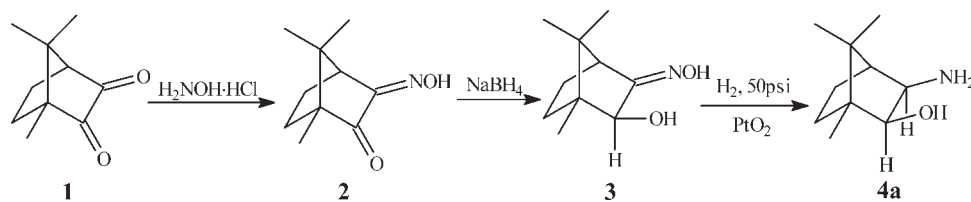
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Received for publication 26 June 2006; Accepted 21 November 2006

DOI: 10.1002/chir.20373

Published online 17 January 2007 in Wiley InterScience (www.interscience.wiley.com).



Scheme 1. Synthesis of chiral camphor-derived β -amino alcohol (**4a**).

aqueous solution of NaOH was poured into the reaction mixture and the solution was extracted with ethyl ether. The ether layer was dried with anhydrous sodium sulfate and concentrated by a rotary evaporator. The crude product was purified by column chromatography with hexane/ethyl acetate (3:1) as a mobile phase to get **4a** in 65% yield. Its characteristics agreed with the data reported in the literature.²³

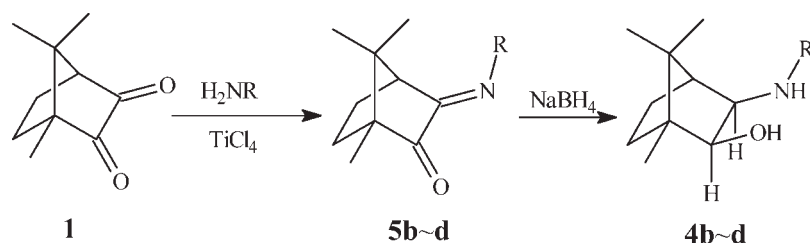
General procedure to chiral N-substituted camphor-derived β -amino alcohols (4b–e**).** To a solution of **4a** (29 mg, 0.2 mmol) and anhydrous sodium sulfate (0.05 g) in 2 mL of anhydrous methanol was added aldehyde or acetone (0.3 mmol). The mixture was stirred in a nitrogen atmosphere at room temperature for 2 h. After the reaction was completed, NaBH₄ (26 mg, 0.7 mmol) was added into the solution and stirred at 0°C for 2 h. Then, the reaction was quenched by titration with HCl aqueous solution until the reaction mixture turned neutral. The solution was extracted with dichloromethane and the extracted solution was dried with anhydrous sodium sulfate and concentrated by a rotary evaporator. The crude product was purified by column chromatography with hexane/ethyl acetate (2:1) as a mobile phase to get the corresponding **4b–e**, and yields were 73, 70, 72, and 68%, respectively. The data obtained for the products agreed with those reported in the literature.^{24,26,27}

RESULTS AND DISCUSSION

As shown in Scheme 4, we synthesized chiral N-substituted camphor-derived β -amino alcohols (**4b–e**), starting from (1*R*)-(+)-camphor, which is commercially avail-

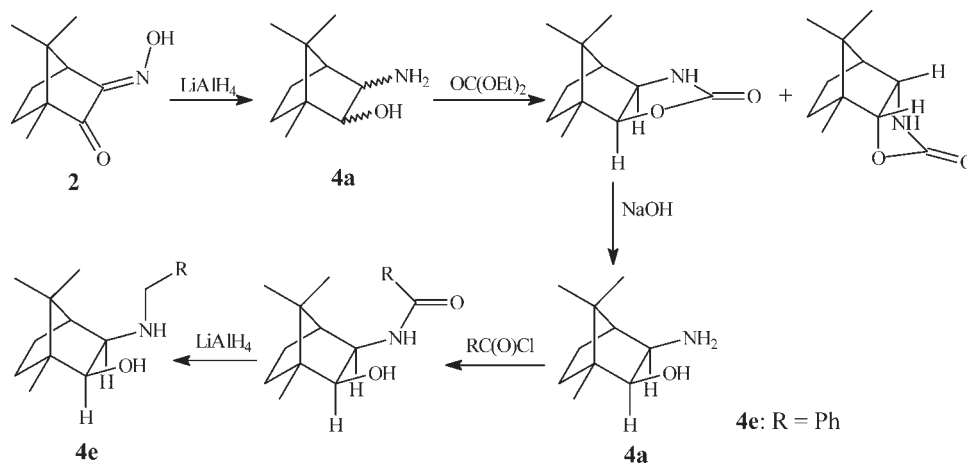
able and cheap. According to Roy and Chakraborti's method,⁴² **2** was prepared by nitrosation of (1*R*)-(+)-camphor. We tried to reduce both carbonyl and oxime functional groups in **2** simultaneously with LiAlH₄ in ether, but the reaction failed to produce the isomerically pure *exo,exo*-amino alcohol (**4a**). This result is consistent with the ones obtained by Thornton²³ and Tanaka²⁵ but contrary to the reports of other groups.^{43–45} As a result, we reduced the two functional groups in **2** sequentially. As Thornton found, NaBH₄ can chemoselectively and stereoselectively reduce the carbonyl group of **2** and leave the oxime group intact, so *exo*-hydroxyoxime (**3**) was prepared according to his method.²³ Thornton carried out stereoselective reduction of the oxime group in **3** with PtO₂ at 50 psi of H₂ gas,²³ but we found it can be carried out stereoselectively and easily with LiAlH₄ in anhydrous ether and *exo,exo*-amino alcohol (**4a**) was generated in 65% yield.

We applied the strategy of reductive amination to N-alkylation of **4a**. First, we condensed **4a** with aldehydes or ketones in methanol. When benzaldehyde was used in the condensation reaction, imino-alcohol (**7**) was not the only product and the other isomer of 1,3-oxazolidine (**6**) was also obtained. However, when acetone, isobutyraldehyde, or butyraldehyde was used in the condensation reactions, 1,3-oxazolidines (**6**) were the only type of products and no traces of imino-alcohols (**7**) were found. Both 1,3-oxazolidines (**6**) and imino-alcohols (**7**) can be reduced by NaBH₄ in methanol at 0°C to obtain the corresponding **4b–e**. Since both the condensation and the hydride reduction reactions use methanol as a solvent, it is plausible to do the two processes of reactions in one pot without isolation of intermediates. The one-pot alkylations of **4a** forming the corresponding chiral N-alkyl-substituted camphor-



4b,5b: R = *n*-Bu, **4c,5c:** R = *i*-C₄H₉, **4d,5d:** R = *i*-C₃H₇

Scheme 2. Synthesis of chiral camphor-derived β -amino alcohols (**4b–d**).

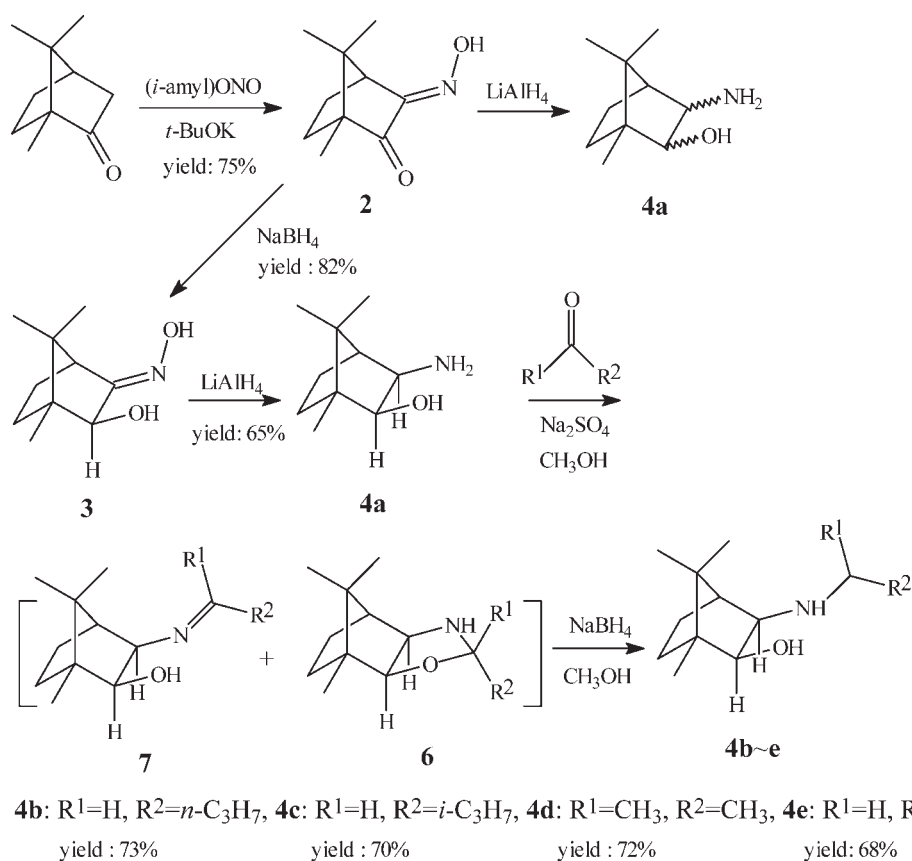


Scheme 3. Synthesis of chiral camphor-derived β -amino alcohol (4e).

derived β -amino alcohols (4b–e) were carried out in around 70% yield.

This alternative route (Scheme 4) from (1*R*)-(+)-camphor to chiral N-substituted camphor-derived β -amino alcohol (4a–e) consists of four steps, has a total yield of 28%,

and uses common and cheap reagents. Advantages of this route over others are (1) cheap and commercially available starting material, (2) few steps to work on, (3) high total yield, and (4) use of common reagents, methods, and apparatuses.



Scheme 4. Synthesis of chiral camphor-derived β -amino alcohols (4a–e).

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